

Learning Objective 1: Proficiency in Human Physiology

	Q1: Describe the physiological mechanisms underlying a specific human organ system (e.g., cardiovascular, respiratory, nervous)	Q2: Explain how changes in physiological parameters (e.g., blood pressure, heart rate) affect organ function and overall homeostasis in the human body.	Q3: List key components of an experiment to investigate a physiological phenomenon in a laboratory setting, including hypothesis formulation, data collection, and interpretation
D	Cardiovascular, Respiratory, Nervous System, Renal, Skeletal Muscular	inc in HB can affect Cardiac output which can adversely effect health.	Key components for an experiment are having a hypothesis that you are testing.
M	For the cardiovascular system, it consists of the heart, blood vessels, and blood. Two mechanisms under the cardiovascular system is the cardiac cycle and and oxygen transport. The cardiac cycle is the heartbeat of a heart, the filling and pumping of blood throughout the heart. The transport of oxygen is through red blood cells that bind within the lungs.	Irregular blood pressure can affect blood vessels. High blood pressure puts strain on blood vessels and can damage them. Low blood pressure may result in not enough oxygen getting to tissues. Heart rate can affect the amount of blood pumped and circulated throughout the body. A decrease in heart rate can also result in not enough oxygen getting into tissues. An increase in heart rate can not allow for proper oxygen diffusion and will also limit the amount of oxygen getting into tissues.	Hypothesis Formulation Experimental Design Data Collection Data Analysis Interpretation
E	The heart, a muscular pump divided into four chambers, propels blood through the body's intricate network of blood vessels. Oxygen-rich blood is delivered to tissues, while waste products are carried away. The electrical rhythm of the heart, initiated by the sinoatrial (SA) node, ensures a steady heartbeat. Arteries, veins, and capillaries facilitate this vital transport, ensuring the body's tissues receive the necessary oxygen and nutrients. The autonomic nervous system, along with reflexes like the baroreceptor reflex, maintains blood pressure, while hormones like angiotensin II regulate blood vessel constriction and dilation.	Changes in these vital parameters will affect the transport of oxygen and nutrients to organs and continuously insufficient supply can ultimately cause organ damage and compromise the organ function. For example, high blood pressure will cause an increased load on the heart, affect the blood vessels in the kidneys, and cause strokes in the brain. Low blood pressure can cause dizziness fainting, and cognitive dysfunction in the brain, and less blood to the organs can cause organ failure.	To investigate lung volume in a lab setting you would use spirometry. The hypothesis formulated may be "If lung volume is measured with spirometry, then the resulting lung volume measured will be approx. X value". To collect data, a spirometer connected to a digital measuring system would be used to obtain values for the breathing. The values collected from the digital output would be used to calculate lung volume by subtracting the volume at the top of the inhale and bottom of the exhale. The resulting volume would be interpreted and compared to the hypothesis. Factors to consider that may affect results could be lung diseases such as asthma that may affect breathing.

Learning Objective 2: Proficiency in Mathematics

	Q1: What techniques would you use to solve a system of ordinary differential equations?	Q2: Explain least squares. Give at least one example of when you would use it?	Q3: What is at least one piece of information you can learn from the eigenvalues of a Linear System?
D	I don't know	- Minimizes squared deviations between observed and predicted values. - Finds optimal line or curve for data.	You can use the eigenvalues to find the stability of the system.
M	I would try to model the system on a computer, by using Euler's method or some other computational method.	least squares method consist of minimizing the sum of squares of gaps between a model and experimental values. I would use it to create a trend that is the representation of experimental values	The eigenvalues of a linear system reveal the stability of its equilibrium points. If the eigenvalues are positive, the system is unstable (solutions grow exponentially). If negative, the system is stable (solutions decay). Zero or complex eigenvalues indicate a more nuanced behavior, such as oscillations or neutral stability.
E	Generally speaking, this depends on whether the ODE is homogeneous or non-homogenous. Homogeneous ODEs can largely be solved by finding eigenvectors and eigenvalues which gets you the general solution. However, for non-homogenous ODEs when there is a homogenous part and the particular part, having a table of laplace transformations can reduce a initial value problem to a matter of mere algebra.	least squares are used to find the best-fitted curve through a given data point. it generally uses the formula $y=mx+c$. it can be used in blood glucose system. where we can track the blood glucose level. here x and y will be blood glucose and readings from the device , m will be the slope and c will be the intercept.	The eigenvalues of a linear system reveal the stability of its equilibrium points. If the eigenvalues are positive, the system is unstable (solutions grow exponentially). If negative, the system is stable (solutions decay). Zero or complex eigenvalues indicate a more nuanced behavior, such as oscillations or neutral stability.

Learning Objective 3: Breadth of Understanding in contemporary BME topics

	Q1: List two recent advancements in biomedical engineering research, technology, and clinical applications across different subfields (e.g., regenerative medicine, bioinformatics, wearable sensors)	Q2: What is the impact of at least one emerging technology (e.g., artificial intelligence, nanotechnology, bioprinting) on the future of biomedical engineering and healthcare	Q3: List at least 2-3 ethical, social, and regulatory issues associated with the development and implementation of new biomedical technologies and interventions
D	Development of bioengineered organs and tissues using 3D bioprinting technology.	AI Improved diagnostics, personalized treatments, and preventative care.	Biased in terms of technology, inequality, patient safety, and health risks either being long term or not for each development.
M	1. Medtronic developed a surgically wearable button which helps in continuous monitoring and early detection of diseases 2. Nano robot which helps in targeting cancer cells	Artificial intelligence greatly impacts the future of biomedical engineering and healthcare. Many patient monitoring sensors and technologies require healthcare providers to analyze collected data. Instead, AI can be trained to interpret the collected data in real time and to notify healthcare providers of concerns or unusual data.	consent from the patients for treatment using new technology and explaining to them the pros and cons of the technology. easy access to these technologies. socially accepting the new technology widely. increasing chances of new employments. regulatory approval from FDA
E	1. 3D Bioprinting: Advancements in 3D bioprinting allow the creation of personalized tissue and organ models for regenerative medicine and drug testing. 2. Wearable Health Devices: Smart wearables, like continuous glucose monitors and ECG patches, provide real-time health data, enhancing remote patient monitoring and personalized treatments in chronic disease management.	Artificial intelligence is one of the most important emerging technologies across many fields right now. It can be very useful for some applications, but relying too much on it might have a negative impact. In MR for example, it can be used to detect patterns in brain tissue that might be linked to certain neurological diseases. However, AI algorithms and data can be subject to nefarious attacks such as "poisoned" data, which can skew results.	Ethical: Issues with privacy and data collection. With AI, data must be used to train the algorithm, but where is the line drawn at what data is available and what is done with the collected data. Social: Issues with rising costs for tech/interventions as it becomes more complex. Insurance coverage and ensuring that treatment availability is not only available to wealthy individuals. Regulatory: Issues with the FDA approval process and ensuring safety but also making sure tech/interventions are available for the public to use. Finding the balance between the two.

Learning Objective 4: Mastery of a specific BME topic

	Q1: List at least 1 area of BME (e.g., medical imaging, devices, neural engineering, tissue engineering, data sciences) that you have pursued in your elective course selections? Provide an overview of thi	Q2: List a few of the key elective courses that you have chosen in your program and how you believe these courses round out your degree in Biomedical Engineering.
D	Medical imaging and data science	Bioinstrumentation: coming from a "medical devices" major in france, I found interesting to continue in this field to strengthen my knowledge. Advanced Biomedical Engineering Design: Same as the other one, I wanted to continue in the devices field.
M	Bioinstrumentation: Bioinstrumentation is a specialized area within biomedical engineering dedicated to the design and development of devices and instruments for measuring biological and physiological processes. This field is essential for diagnosing diseases, monitoring health conditions, and facilitating research in life sciences. Some key aspects will be the measurement techniques for example or the data processing.	I have taken Quantitative neural function which helped me in understanding the bridging engineering and neuroscience. The lab part taught me how simulation software can be used to simulate functions of body like heartbeat etc.
E	Medical Imaging is one area I have pursued so far. This area is very important, and the technology or methods used are changing constantly. Generally, it involves imaging some part of the body, and then uses a model or method of some sort to interpret the data and obtain images that can be used to diagnose a patient by a doctor. I have learned the most about MRI, which uses magnets to align protons in the body and reads their signal as they decay (brief explanation). Some key aspects of this area are the emerging use of AI and low-field MR.	BIOL 550 - Bioinformatics: This course is expanding my knowledge as I am learning Linux and gaining more applicable CS knowledge. BME 535 - Magnetic Resonance Imaging: This course helped me become a better researcher, and I learned more about current methods in MRI. BME 560 - Methods in Biomedical Data Science: This course gave me knowledge about how statistics is used in Machine Learning, specifically on biological data.